(+65) 9812 2021
longhp1618@gmail.com

Long P. Hoang

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longhp1618.github.io

RESEARCH INTEREST

Long's research focuses on advancing and interpreting multi-objective deep neural networks. He specifically investigates techniques to improve model helpfulness, and safety simultaneously. A key area of his work involves scaling these effectiveness measures to large-scale applications, particularly Large Language Models (LLMs).

EDUCATION

PhD **Singapore University of Technology and Design (SUTD)** **09/2024 – Present**

- Academic advisors: Asst. Prof. Wenxuan Zhang and Prof. Wei Lu.
- Major: Artificial Intelligence.

Bachelor **Hanoi University of Science and Technology (HUST)** **08/2018 – 09/2022**

- Academic advisor: Dr. Thang N. Tran
- Major: Mathematics and Informatics. CPA: 3.11/4.00 (Rank 18 out of 94 in his department).

SELECTED PUBLICATIONS

[1] **Long P. Hoang**†, Hai V. Le†, Shaoyang Xu, Wei Lu, Wenxuan Zhang, Safety Paradox: How Enhanced Safety Awareness Leaves LLMs Vulnerable to Posterior Attack, *In Proceedings of EMNLP*, 2026 (Oral).

[2] **Long P. Hoang**, Wenxuan Zhang, Wei Lu, Difficulty–Diversity Collaborative Filtering for Data-Efficient LLM Fine-Tuning, *In Proceedings of ICLR*, 2026.

[3] Tuan A. Tran†, **Long P. Hoang**†, Dung D. Le, Thang N. Tran, A Framework for Controllable Pareto Front Learning with Completed Scalarization Functions and its Applications, *Neural Networks* , 2024.

[4] **Long P. Hoang**, Dung D. Le, Tuan A. Tran, Thang N. Tran, Improving Pareto Front Learning via Multi-Sample Hypernetworks, *In Proceedings of AAAI*, 2023 (Oral).

PREPRINTS

[1] **Long P. Hoang**, Yiran Zhao, Wei Lu, Wenxuan Zhang, Multilingual Fine-Tuning via Localized Gradient Conflict Resolution, *arXiv preprint arXiv:2606.05613*, 2026.

EXPERIENCE

PhD Student **The inclusive NLP Lab**

Advisor: Prof. Wei Lu and Asst. Prof. Wenxuan Zhang

How Enhanced Safety Awareness Leaves LLMs Vulnerable to Posterior Attack *10/2025 – 05/2026*

- Developed Posterior Attack, a single-query jailbreak method that demonstrates high efficacy against a wide range of LLMs, including GPT-5 and Claude 4.6.
- Using Bayesian inference, developed a theoretical framework showing the Safety Paradox between LLM internal safety-awareness and its vulnerability under Posterior Attack (accepted to EMNLP 2026).

†Co-First Author

Multilingual Fine-Tuning via Localized Gradient Conflict Resolution *01/2026 – 05/2026*

- Formulated multilingual fine-tuning as a multi-objective optimization (MOO) problem, then developed distributed gradient-based MOO training to improve LLM performance up to 4% (submitted to ICLR 2027).

Difficulty-Diversity Collaborative Filtering for Data-Efficient LLM Fine-Tuning *04/2025 – 12/2025*

- Developed a collaborative filtering framework to select a small, challenging, and diverse dataset personalized for a given language model (accepted to ICLR 2026).
- Empirically validated that fine-tuning with 1,000 high-quality examples selected via our method yields better performance than using the entire 220,000-example dataset.

Research Assistant **College of Engineering and Computer Science, VinUniversity**

Advisor: Asst. Prof. Dung D. Le

Expensive Multi-Objective Optimization *03/2023 – 10/2023*

- Approximated the entire trade-off curve of black-box objects using Pareto Front Learning with Hypernetworks computed by Gaussian Processes (accepted to AAAI 2025).

Profiling the Pareto Front in Multi-Task Learning *02/2022 – 10/2022*

- Proposed a novel method named Multi-Sample Hypernetwork to approximate the entire trade-off curve of conflicting objectives (accepted to AAAI 2023).

Undergraduate Research Assistant **School of Applied Mathematics and Informatics, HUST**

Advisor: Dr. Tran Ngoc Thang

Multi-Objective Optimization with Completed Scalarizations *02/2022 – 02/2023*

- Proposed and proved the convergence of a new framework for Pareto Multi-Task Learning with Scalarization Functions in the pseudo-convex and quasiconvex assumptions (accepted to the journal Neural Networks).

ADWARDS & CERTIFICATES

- AISG PhD Fellowship, granted by AI Singapore, Sep 2024 – Sep 2028.
- Certificate of Completion of Developer Circles Vietnam Innovation Challenge in Data Science, sponsored by Facebook, Nov 2020.

REFERENCES

1. **Assistant Professor Wenxuan Zhang** (Ph.D), Information Systems Technology and Design, Singapore University of Technology and Design *wxzhang@sutd.edu.sg*
2. **Professor Wei Lu** (Ph.D), College of Computing and Data Science, Nanyang Technological University *wei.lu@ntu.edu.sg*
3. **Assistant Professor Dung D. Le** (Ph.D), College of Engineering and Computer Science, VinUniversity *dung.ld@vinuni.edu.vn*
4. **Dr. Thang N. Tran** (Ph.D), Faculty of Applied Mathematics and Informatics, Hanoi University of Science and Technology *thang.tranngoc@hust.edu.vn*